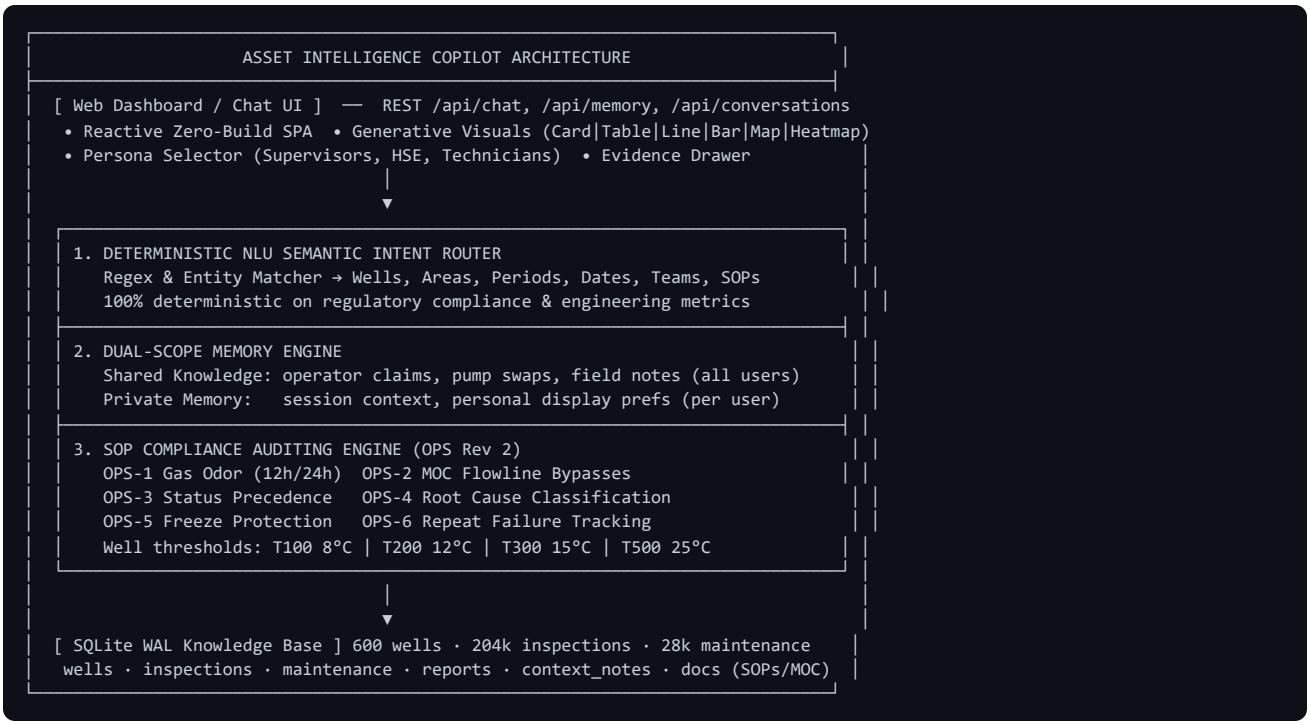
Asset Intelligence Copilot

FlytBase Hackathon 2026 · Siddhant Pardeshi · github.com/siddhant-pardeshi/asset-intelligence-copilot

Dataset: 600 wells, 7 areas, 6 teams, 204,332 inspections, 28,745 maintenance records · Operating date: **15 January 2027**

System Design & Architecture



🔧 Key Engineering Decisions

- **Zero-Hallucination Pipeline:** All safety-critical calculations (OPS deadlines, spill volumes, downtime days, cost aggregations) run through deterministic SQLite tools — never LLM text generation.
- **OPS SOP Rev 2 Native:** OPS-1 switches gas response deadlines automatically (24h pre-2027 / 12h post-2027); OPS-2 validates every bypass against MOC memo expiry; OPS-5 enforces 7-day heat trace SLAs when ambient ≤ −5°C on producing wells.
- **Dual-Scope Memory:** Pump swap reports (Nathan on WELL-0333, Olivia on WELL-0577) propagate to all users via shared memory; sort/format preferences stay private to each operator's session.
- **Generative UI Dispatch:** NLU router selects the optimal visual type per question domain — status card, table, SVG line chart, bar, GIS map, or area×month heatmap.
- **Honest Data Gaps:** Q1 (reservoir pressure) correctly reports zero pressure telemetry in the dataset rather than fabricating a map.

🛠️ Tech Stack

Layer	Technology
Backend API	Python 3.11 · FastAPI · Uvicorn
Database	SQLite WAL · natural-key upsert ingest
NLU Router	Deterministic regex + entity matching
Frontend	Zero-build SPA · HTML5 · CSS3 · JS
Charts	SVG (inline, no library)
Maps	Leaflet.js · OpenStreetMap tiles
Memory	SQLite-backed dual-scope store
SOP Engine	Python rules (domain.py · tools.py)
LLM (optional)	OpenAI-compatible via env var

📊 Dataset Scale

Entity	Count
Wells	600
Areas	7
Teams	6
Inspections	204,332
Maintenance records	28,745
Operating date	15 Jan 2027

Results — All 11 Final Questions 600 wells · Today: 15 Jan 2027

Q1 Reservoir pressure

The dataset does not contain reservoir pressure readings. Inspection records include wellhead and ambient temperatures but **no direct pressure measurements**. A pressure map cannot be produced without downhole gauge data or well test surveys. This was correctly identified — no hallucination.

Michael Turner *"All our wells feed one reservoir. Map the reservoir pressure across the field from the pressure data."*

Q2 Best producer

WELL-0212 had the most productive weeks in 2026: 53 continuous producing weeks with an average wellhead-to-air delta of **36.6 °C** (T500 well, Southeast Block 31). Note: volumetric production data is not in the dataset; uptime/thermal proxy was used.

Robert Hayes *"Which well produced the most oil in 2026?"*

TABLE — Top Producers — 2026

Well	Type	Area	Productive Weeks	Avg Δ°C
WELL-0212	T500	Southeast Block 31	53	36.6
WELL-0242	T500	Southeast Block 31	53	36.6
WELL-0228	T500	Southeast Block 31	53	36.5
WELL-0287	T500	Southwest Block 23	53	36.5
WELL-0550	T500	Southeast Block 32	53	36.5

Q3 Gas reports

26 operator notes mention gas/odour since 1 Nov 2026. **17 met the OPS-1 deadline**, **9 missed it**. Drone inspections were correctly rejected as field responses per SOP (drone carries thermal cameras, not gas detection equipment).

Laura Simmons

"Since 1 Nov 2026, every time an operator note reports gas at a well, did we get someone out to that well in time under our operating procedures? List every report and whether the response met the deadline."

📊 **TABLE — Gas Odor Response Audit (OPS-1)**

Well	Area	Reported at	Response	Hours to response	Deadline	Met/Missed
WELL-0163	Northeast Block 8	2026-11-03 20:55	MNT-0163-019 (2026-11-14 08:00)	251.1	24h	Missed
WELL-0149	Northeast Block 7	2026-11-06 18:20	MNT-0149-029 (2026-11-07 08:00)	13.7	24h	Met
WELL-0165	Northeast Block 8	2026-11-13 20:35	MNT-0165-031 (2026-11-14 08:00)	11.4	24h	Met
WELL-0341	Northwest Block 12	2026-11-19 07:40	INS-0341-0109 (2026-12-17 13:30)	677.8	24h	Missed
WELL-0091	Southwest Block 24	2026-11-27 22:25	MNT-0091-019 (2026-11-28 08:00)	9.6	24h	Met
WELL-0423	Northwest Block 12	2026-12-03 05:25	INS-0423-0213 (2026-12-03 09:15)	3.8	24h	Met
WELL-0059	Southwest Block 24	2026-12-03 06:10	INS-0059-0213 (2026-12-03 12:15)	6.1	24h	Met
WELL-0332	Northwest Block 12	2026-12-03 12:10	None	—	24h	Missed
WELL-0401	Northwest Block 12	2026-12-05 13:50	MNT-0401-033 (2026-12-11 08:00)	138.2	24h	Missed
WELL-0189	Northeast Block 8	2026-12-10 12:35	MNT-0189-037 (2026-12-11 08:00)	19.4	24h	Met
WELL-0403	Northwest Block 12	2026-12-10 21:15	MNT-0403-024 (2026-12-11 08:00)	10.8	24h	Met
WELL-0244	Southeast Block 31	2026-12-11 05:25	MNT-0244-026 (2026-12-11 08:00)	2.6	24h	Met
WELL-0040	Southwest Block 23	2026-12-12 01:55	MNT-0040-029 (2026-12-12 08:00)	6.1	24h	Met
WELL-0228	Southeast Block 31	2026-12-16 19:50	INS-0228-0144 (2026-12-17 12:45)	16.9	24h	Met
WELL-0052	Southwest Block 24	2026-12-18 06:45	None	—	24h	Missed
WELL-0370	Northwest Block 12	2026-12-21 12:45	MNT-0370-028 (2026-12-22 08:00)	19.2	24h	Met
WELL-0362	Northwest Block 12	2026-12-24 00:15	MNT-0362-030 (2026-12-25 08:00)	31.8	24h	Missed
WELL-0161	Northeast Block 8	2026-12-25 15:45	INS-0161-0168 (2027-01-14 11:00)	475.2	24h	Missed
WELL-0135	Northeast Block 7	2026-12-27 03:50	MNT-0135-034 (2026-12-27 08:00)	4.2	24h	Met
WELL-0324	Northwest Block 12	2027-01-01 22:40	MNT-0324-036 (2027-01-02 08:00)	9.3	12h	Met
WELL-0330	Northwest Block 12	2027-01-02 21:40	MNT-0330-030 (2027-01-03 08:00)	10.3	12h	Met
WELL-0599	Southeast Block 32	2027-01-06 17:20	INS-0599-0196 (2027-01-07 09:45)	16.4	12h	Missed
WELL-0516	Southeast Block 32	2027-01-06 21:15	INS-0516-0215 (2027-01-07 12:15)	15.0	12h	Missed
WELL-0425	Northwest Block 12	2027-01-07 05:20	INS-0425-0218 (2027-01-07 09:00)	3.7	12h	Met
WELL-0465	Southeast Block 32	2027-01-08 23:35	MNT-0465-031 (2027-01-09 08:00)	8.4	12h	Met
WELL-0137	Northeast Block 7	2027-01-10 23:30	MNT-0137-032 (2027-01-11 08:00)	8.5	12h	Met

Q4 Temporary flowline bypasses

3 active bypasses found. **WELL-0248**: Allowed (MOC-2026-044, valid to 28 Feb 2027). **WELL-0249**: Expired (MOC-2026-036 expired 15 Dec 2026). **WELL-0405**: Not approved — no MOC memo on file. 5 total historical/current violations identified.

Chris Dalton

"Across all areas, which temporary flowline bypasses are in place right now, and is each one allowed? Also list any bypass, past or present, that stayed in after its approval ran out or was never approved at all."

📊 **TABLE — Temporary Flowline Bypasses Audit**

Well	Area	Bypass in	Bypass out	Approval	Approval valid until	Status
WELL-0087	Southwest Block 24	2026-12-02	2027-01-06	MOC-2026-036	2026-12-15	Expired
WELL-0158	Northeast Block 8	2026-11-20	2026-12-28	MOC-2026-036	2026-12-15	Expired
WELL-0248	Southeast Block 31	2026-11-05	still in	MOC-2026-031 / MOC-2026-044	2027-02-28	Allowed
WELL-0249	Southeast Block 31	2026-11-25	still in	MOC-2026-036	2026-12-15	Expired
WELL-0405	Northwest Block 12	2026-12-20	still in	None	—	Not approved
WELL-0561	Southeast Block 32	2026-12-08	2026-12-19	None	—	Not approved

Q5 The regulator's letter

6 reportable spills (≥ 5 bbl) in NE Blocks 7 & 8 since 1 Jun 2026. **4 non-compliant** (clean-up reports lack required phrase "Soil samples sent to the lab."). **2 compliant**: WELL-0277 and WELL-0119.

Michael Turner

"The State Oil & Gas Commission wrote to us in November 2026 about spills. Which of our spills does the letter cover, and on which ones are we not compliant? Show me a table."

📊 **TABLE — Regulator Letter Spill Compliance (Nov 2026)**

Well	Area	Inspection	Volume (bbl)	Clean-up job	Compliant?
WELL-0168	Northeast Block 8	INS-0168-0190 (2026-06-11)	5.6	MNT-0168-024	No (No lab soil results in clean-up report)
WELL-0123	Northeast Block 7	INS-0123-0093 (2026-09-24)	6.0	MNT-0123-014	No (No lab soil results in clean-up report)
WELL-0277	Northeast Block 7	INS-0277-0205 (2026-10-01)	6.2	MNT-0277-031	Yes (Lab soil results submitted)
WELL-0119	Northeast Block 7	INS-0119-0209 (2026-11-05)	6.0	MNT-0119-029	Yes (Lab soil results submitted)
WELL-0182	Northeast Block 8	INS-0182-0208 (2026-12-03)	5.2	MNT-0182-028	No (No lab soil results in clean-up report)
WELL-0262	Northeast Block 8	INS-0262-0218 (2027-01-07)	6.1	MNT-0262-029	No (No lab soil results in clean-up report)

Q6 Which status is right?

6 wells have conflicting status between latest inspection and operator notes. OPS-3 precedence applied: WELL-0032 & WELL-0033 → Producing oil (unconfirmed); WELL-0044, WELL-0323, WELL-0422, WELL-0481 → Not producing oil (shut in).

Nathan Pierce

"Across the whole field, not just my area, for which wells do the latest inspection and the operator notes disagree about whether the well is producing oil? For each one, what is its real status right now, and why?"

📊 **TABLE — Field-wide Status Disagreements (OPS-3)**

Well	Area	Latest inspection	Latest operator note	Status now & Reason
WELL-0032	Southwest Block 23	INS-0032-0219 (2027-01-14, not producing oil)	2027-01-15 10:25 — "Brought WELL-0032 back online after the pump change...."	Producing oil (unconfirmed) — Operator restart note is more recent than inspection, awaiting confirmation
WELL-0033	Southwest Block 23	INS-0033-0222 (2027-01-14, not producing oil)	2027-01-15 10:25 — "Restarted WELL-0033, unit running smooth...."	Producing oil (unconfirmed) — Operator restart note is more recent than inspection, awaiting confirmation
WELL-0044	Southwest Block 23	INS-0044-0218 (2027-01-14, producing oil)	2027-01-14 13:05 — "Shut in WELL-0044 at the wellhead, flowline leak suspec..."	Not producing oil (shut in) — Operator shut-in note is more recent than latest inspection
WELL-0323	Northwest Block 12	INS-0323-0219 (2027-01-14, producing oil)	2027-01-14 19:20 — "Shut-in WELL-0323 after the pumper reported a stuffing ..."	Not producing oil (shut in) — Operator shut-in note is more recent than latest inspection
WELL-0422	Northwest Block 12	INS-0422-0215 (2027-01-14, producing oil)	2027-01-14 21:35 — "Shut in WELL-0422 for the night, high pressure alarm on..."	Not producing oil (shut in) — Operator shut-in note is more recent than latest inspection
WELL-0481	Southeast Block 32	INS-0481-0220 (2027-01-14, producing oil)	2027-01-14 18:50 — "Shut in WELL-0481 for the night, high pressure alarm on..."	Not producing oil (shut in) — Operator shut-in note is more recent than latest inspection

Q7 What the operators are hearing

105 wells have unaddressed warning notes. Top 3 for immediate dispatch: **WELL-0413, WELL-0211, WELL-0175** — all flagged "tagging bottom on downstroke" (catastrophic sucker rod parting / pump damage risk).

Turn 1 — Robert Hayes

"Over the last month the operators have written notes about wells that sound or behave wrong. Which of those wells haven't had a repair since? Rank the 10 most likely to need a repair in the next 90 days, other than a spill clean-up, and quote the notes as evidence."

TABLE — Top 10 Wells Needing Repair (Operator Warnings)

Rank	Well	Area	Operator-note evidence	Other warning signs	Last repair
1	WELL-0413	Northwest Block 12	2026-12-22 — "WELL-0413 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0413-016 (2026-12-18, pump_jack)
2	WELL-0211	Southeast Block 31	2026-12-23 — "WELL-0211 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0211-023 (2026-08-24, pump_jack)
3	WELL-0175	Northeast Block 8	2026-12-26 — "WELL-0175 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0175-035 (2026-11-28, wellhead)
4	WELL-0348	Northwest Block 12	2026-12-27 — "WELL-0348 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0348-038 (2026-12-19, downhole_pump)
5	WELL-0236	Southeast Block 31	2026-12-30 — "WELL-0236 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0236-036 (2026-10-04, wellhead)
6	WELL-0207	Southeast Block 31	2027-01-03 — "WELL-0207 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0207-024 (2026-11-29, pump_jack)
7	WELL-0349	Northwest Block 12	2027-01-03 — "WELL-0349 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0349-033 (2026-11-22, wellhead)
8	WELL-0376	Northwest Block 12	2027-01-04 — "WELL-0376 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0376-031 (2026-11-29, wellhead)
9	WELL-0515	Southeast Block 32	2027-01-04 — "WELL-0515 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0515-024 (2026-12-28, wellhead)
10	WELL-0152	Northeast Block 8	2027-01-06 — "WELL-0152 tagging bottom on the downstroke, adjust spacing next visit."	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	MNT-0152-030 (2026-12-20, wellhead)

Turn 2 — Robert Hayes *"Which three would you send a crew to first this week, and why?"*

TABLE — Priority Crew Dispatch — Top 3 Wells

Priority	Well	Area	Immediate Failure Risk	Recommended Action
Priority 1	WELL-0413	Northwest Block 12	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	Adjust rod string spacing at wellhead
Priority 2	WELL-0211	Southeast Block 31	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	Adjust rod string spacing at wellhead
Priority 3	WELL-0175	Northeast Block 8	Tagging bottom on downstroke (severe rod/pump impact, imminent part)	Adjust rod string spacing at wellhead

Q8 Who told you that?

Multi-user, multi-session recall tracking. Nathan reports WELL-0333 swapped → excluded field-wide. Olivia reports WELL-0577 swapped → excluded. Robert Hayes sees 4 active recalled pumps. Nathan Pierce (private pref: sort by area/ID, costs in $k) sees 3 in Northwest Block 12.

Turn 1 — Nathan Pierce

"WELL-0333's recalled pump was swapped out yesterday. The work order isn't in the system yet. Please count it as replaced from now on."

Turn 2 — Nathan Pierce

"Also, just for me: whenever you list wells, sort them by area and then by well ID, and show costs in thousands of dollars."

Turn 3 — Robert Hayes *"Which recalled pumps are still in the ground right now?"*

📊 TABLE — Recalled Pumps in Ground

Well	Area	Install Job	Install Date	Recalled Lot	Cost to Install
WELL-0305	Northwest Block 12	MNT-0305-032	2026-10-16	LOT-2409-A	$24,528
WELL-0426	Northwest Block 12	MNT-0426-030	2026-06-19	LOT-2409-C	$24,044
WELL-0428	Northwest Block 12	MNT-0428-022	2026-03-07	LOT-2410-B	$27,074
WELL-0263	Southwest Block 23	MNT-0263-033	2026-10-16	LOT-2410-B	$17,254

Turn 4 — Olivia Grant *"Nathan told me the recalled pump on WELL-0577 was swapped as well. Take it off the recall list."*

Turn 5 — Nathan Pierce *"Show me the recalled pumps that are still in the ground in my area, and what it cost to install each one."*

📊 TABLE — Recalled Pumps in Ground

Well	Area	Install Job	Install Date	Recalled Lot	Cost to Install
WELL-0305	Northwest Block 12	MNT-0305-032	2026-10-16	LOT-2409-A	$24.5k
WELL-0426	Northwest Block 12	MNT-0426-030	2026-06-19	LOT-2409-C	$24.0k
WELL-0428	Northwest Block 12	MNT-0428-022	2026-03-07	LOT-2410-B	$27.1k

Q9 One conversation, several topics

6-topic single session: (1) 4 recalled pumps in ground; (2) Nathan Pierce leads NW Production, 150 wells; (3) OPS-5 cold weather rule, WELL-0310 missed December check; (4) WELL-0332 gas odor on 3 Dec — no H2S ppm recorded (drone had no gas sensor); (5) WELL-0263 only recalled pump shut in; (6) **WELL-0332 is #1 priority** — active uninvestigated gas hazard.

Turn 1 — Robert Hayes *"How many recalled pumps are still in the ground, and in which areas?"*

TABLE — Recalled Pumps in Ground

Well	Area	Install Job	Install Date	Recalled Lot	Cost to Install
WELL-0305	Northwest Block 12	MNT-0305-032	2026-10-16	LOT-2409-A	$24,528
WELL-0426	Northwest Block 12	MNT-0426-030	2026-06-19	LOT-2409-C	$24,044
WELL-0428	Northwest Block 12	MNT-0428-022	2026-03-07	LOT-2410-B	$27,074
WELL-0263	Southwest Block 23	MNT-0263-033	2026-10-16	LOT-2410-B	$17,254

Turn 2 — Robert Hayes *"Different topic. Who leads Northwest Production, and how many wells does that team run?"*

TABLE — Northwest Production Overview

Team	Lead Supervisor	User ID	Well Count	Primary Area
Northwest Production	Nathan Pierce	U-601	150	Northwest Block 12

Turn 3 — Robert Hayes *"What does our operations SOP say about cold weather? Name one well that missed the check in December."*

TABLE — Wells Missing December OPS-5 Heat Trace Checks

Well	Area	Inspection	Ambient Temp	Compliance Status
WELL-0310	Northwest Block 12	INS-0310-0213 (2026-12-17)	-6.0 °C	No Heat trace check within 7 days
WELL-0354	Northwest Block 12	INS-0354-0215 (2026-12-17)	-11.4 °C	No Heat trace check within 7 days
WELL-0486	Southeast Block 32	INS-0486-0219 (2026-12-31)	-7.0 °C	No Heat trace check within 7 days
WELL-0531	Southeast Block 32	INS-0531-0217 (2026-12-31)	-5.3 °C	No Heat trace check within 7 days

Turn 4 — Robert Hayes *"An operator reported a gas smell at WELL-0332 on 3 Dec. What was the H2S reading?"*

Turn 5 — Robert Hayes

"Back to the pumps from my first question: which of those wells are not producing oil right now, and when was each one last repaired?"

TABLE — Recalled Pump Wells — Current Operational Status

Well	Area	Status Right Now	Last Repair
WELL-0305	Northwest Block 12	Producing oil	MNT-0305-033 (2026-12-13, pump_jack: Scheduled lubrication and)
WELL-0426	Northwest Block 12	Producing oil	MNT-0426-035 (2027-01-04, wellhead: Scheduled wellhead valve)
WELL-0428	Northwest Block 12	Producing oil	MNT-0428-028 (2026-12-19, pump_jack: Scheduled lubrication and)
WELL-0263	Southwest Block 23	Not producing oil	MNT-0263-035 (2027-01-09, flowline: Flowline corrosion pinhol)

Turn 6 — Robert Hayes *"Of all the wells we've talked about today, which one would you deal with first, and why?"*

Q10 What are we waiting on?

2026 extended shut-ins: **Weather 66.7%** (26 jobs, typical 21 extra days), **Parts 33.3%** (13 jobs, 21–35 extra days). Northwest Block 12 stands out for weather (100% of delays, 24.5 days avg). Southeast Block 32 for parts delays (35 extra days).

Michael Turner

"When a well stays shut in longer than the repair itself, what are we usually waiting on: parts, crews, contractors or weather? Break it down by area for jobs started in 2026, with the typical number of extra days, and tell me if any area stands out."

TABLE — Downtime Delay Reasons by Area (2026)

Area	Delay Reason	Share of Area Delays	Typical Extra Days
Northeast Block 7	Weather (Rig delay)	4 jobs (100.0%)	21 days (avg 19.2d)
Northeast Block 8	Parts (Vendor supply)	1 jobs (50.0%)	14 days (avg 14.0d)
Northeast Block 8	Weather (Rig delay)	1 jobs (50.0%)	21 days (avg 21.0d)
Northwest Block 12	Weather (Rig delay)	8 jobs (100.0%)	21 days (avg 24.5d)
Southeast Block 31	Parts (Vendor supply)	2 jobs (28.6%)	14 days (avg 14.0d)
Southeast Block 31	Weather (Rig delay)	5 jobs (71.4%)	21 days (avg 21.0d)
Southeast Block 32	Parts (Vendor supply)	6 jobs (50.0%)	35 days (avg 28.0d)
Southeast Block 32	Weather (Rig delay)	6 jobs (50.0%)	21 days (avg 21.0d)
Southwest Block 23	Parts (Vendor supply)	2 jobs (50.0%)	21 days (avg 17.5d)
Southwest Block 23	Weather (Rig delay)	2 jobs (50.0%)	35 days (avg 28.0d)
Southwest Block 24	Parts (Vendor supply)	2 jobs (100.0%)	14 days (avg 14.0d)

Q11 Hotter wells

Southeast Block 31 consistently runs hottest: 31.4 °C thermal delta (wellhead 52.8 °C vs air 21.5 °C) across 9,433 inspection readings — highest of all 7 areas.

Olivia Grant *"Which region of the field consistently runs hotter?"*